

Final Report

Thoa Thanh Thi Le

1. Abstract

2. Introduction

Total fertility rates (TFRs) are key population indicators that are widely monitored because they influence age structure and population growth (Wilmoth et al.2022; World Health Organisation). Globally, fertility rates have declined over recent decades (Chen et al. 2018; Shair et al. 2024; Vanella 2024). In Singapore, the total fertility rate has dropped below the replacement level required for a population to replace itself across generations (Jones & Hamid 2015; Tan, 2024; Wilmoth et al. 2022). Therefore, forecasting TFR and total live births (TLB) may provide insight into future fertility patterns and population change in Singapore. Preliminary exploratory data analysis indicated a long-term decline in TFR and TLB with periodic fluctuations, motivating the use of forecasting models capable of capturing these patterns.

Data from 1960 to 2024 on TLB and TFR were obtained from the Singapore Department of Statistics (2023). The research question addressed in this report is: Can ARIMA models accurately forecast Singapore's TLB and TFR while capturing short-term fluctuations?

This report is structured as follows. Part 3 describes the fertility decline in Singapore, Part 4

3. Literature Review

Around 1965, Singapore introduced policies aimed at reducing population growth, which contributed to subsequent declines in fertility rates (Chen et al. 2018; Goodkind 1996; Singh et al. 1988; Vanella 2024). Despite this long-term decline, there may be cyclical patterns in TFR and TLB. These fluctuations may be associated with cultural preferences surrounding Lunar zodiac Dragon years (1964, 1976, 1988, 2000, 2012 and 2024), which are considered especially auspicious years and have been associated with higher fertility rates and live births (Goodkind 1996; Jones & Hamid 2015). Forecasting such series therefore requires models that can capture both long-term trends and short-term variation. Among the various models for time series forecasting, ARIMA models are particularly flexible because they allow selection from a broad class of models rather than imposing a single model form on every time series (Kolambe & Arora 2024; Newbold 1983). Consequently, ARIMA models were selected as the forecasting approach for Singapore's TFR and TLB.

Methods

For both TFR and TLB, an identical modelling workflow was applied. The analysis focused on the period from 1960-2012 for model fitting, with observations 2013-2024 reserved for evaluating forecast accuracy. Initial plots were used to examine the presence of trends, seasonality, and other notable patterns, which indicated a long-term decreasing trend in both series. Several transformations were considered including logarithmic transformation and first- and second-order differencing. Stationarity was assessed using the

KPSS test. For each transformed series, autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to find candidate ARIMA models. The candidate ARIMA models were then fitted and the residual ACF and PACF were examined to assess whether the residuals resembled white noise. Forecast accuracy was evaluated using mean square error (MSE) and mean absolute error (MAE) while model selection was guided by Akaike's Information Criterion (AIC) and the corrected Akaike's Information Criterion (AICc). This procedure was applied consistently to both series to ensure comparability of results.

Figure 1. Initial Plot of Total Fertility Rate (TFR)

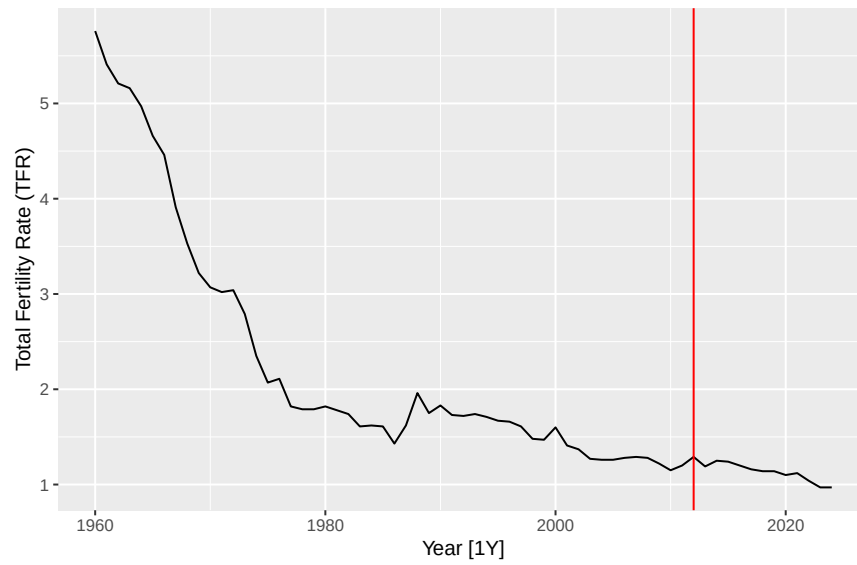
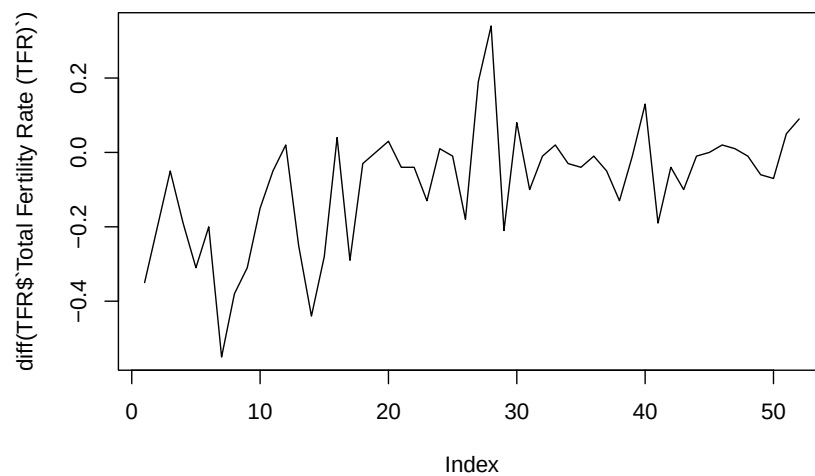
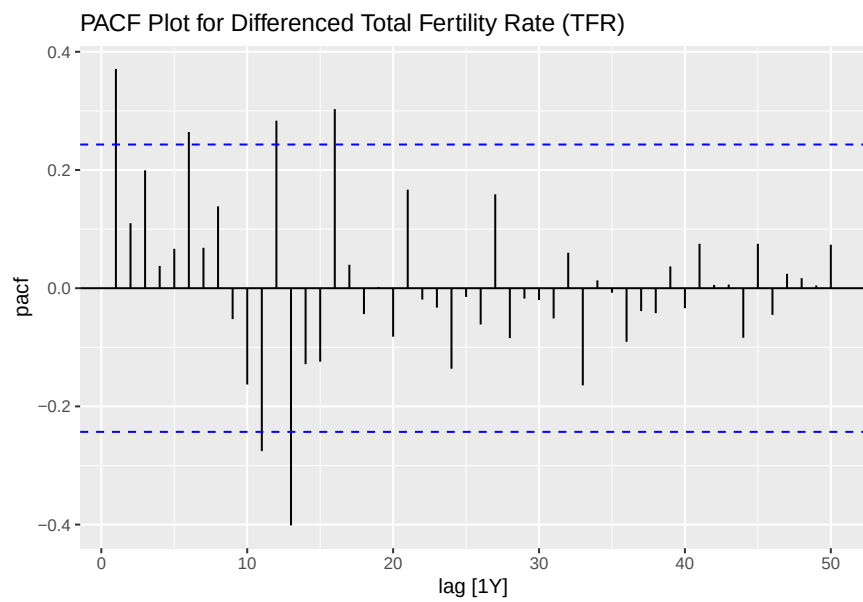
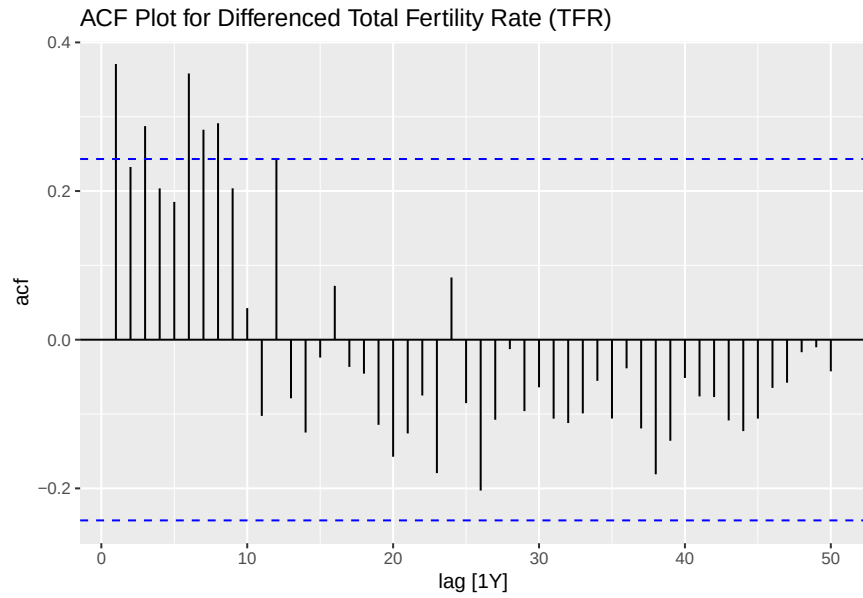


Figure 1 show a long-term decreasing trend over time

Plot of Differenced Total Fertility Rate (TFR) until 2012

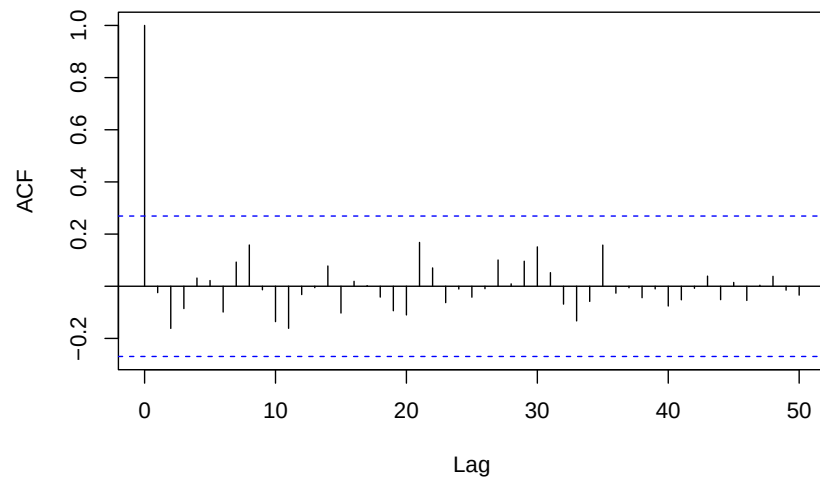


```
##      kpss_stat kpss_pvalue
##      0.8412374  0.0100000
```

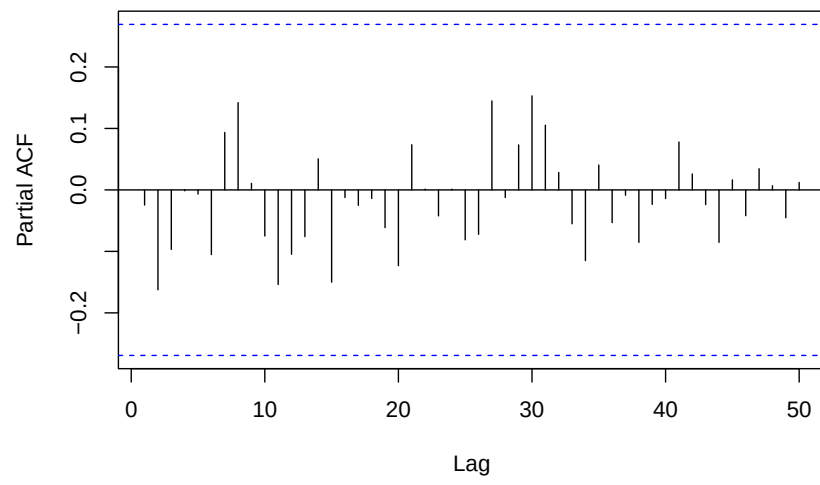


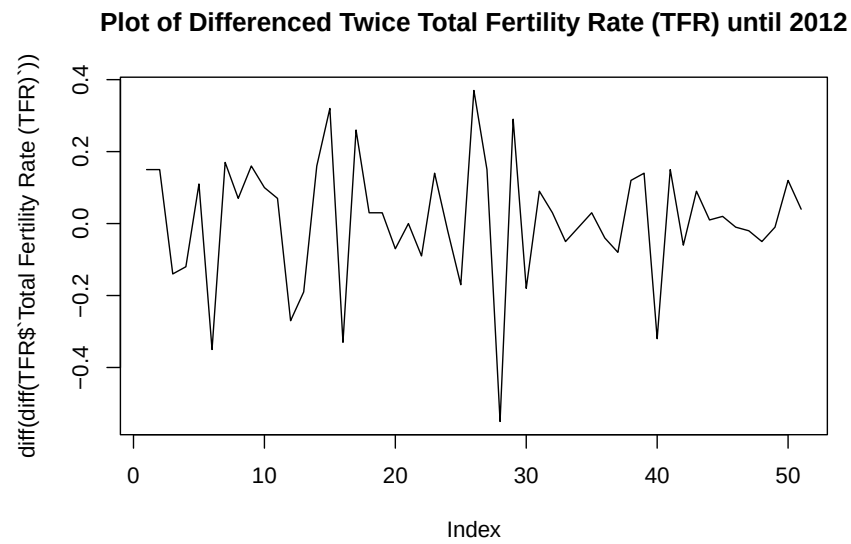
```
## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.3647 -0.1061  0.4577 -0.2586  0.2984  0.2452  0.2006  0.1578 -0.0293
## s.e.  0.1240  0.1289  0.1246  0.1128  0.0935  0.0832  0.0952  0.0928  0.0996
##      ar10     ar11     ar12     ar13     ar14     ar15     ar16
##      -0.0306 -0.3594  0.3889 -0.6048 -0.0419 -0.3361  0.5135
## s.e.  0.0973  0.0913  0.1039  0.1066  0.1350  0.1295  0.1397
##
## sigma^2 = 0.009716: log likelihood = 48.99
## AIC=-63.99  AICc=-45.99  BIC=-30.82
```

ACF of Residuals for ARIMA(16,1,0) for TFR

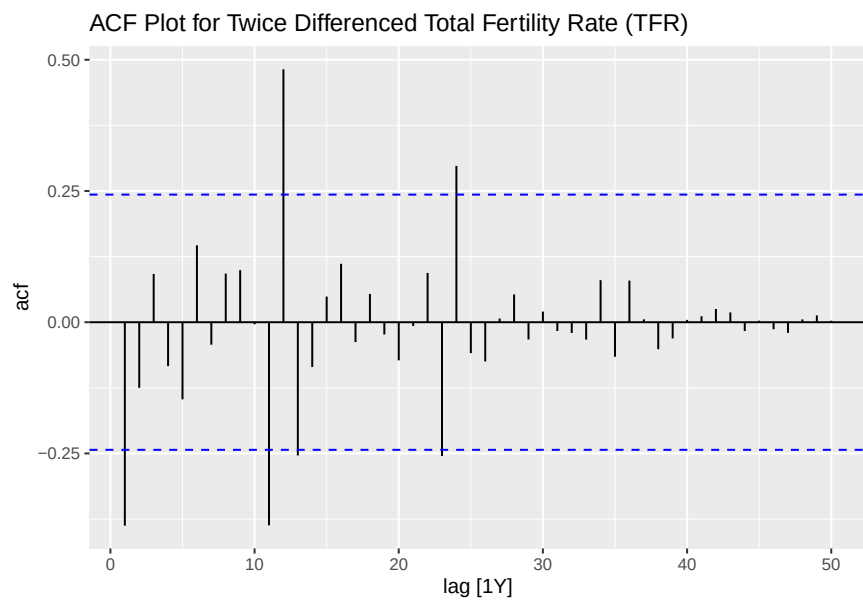


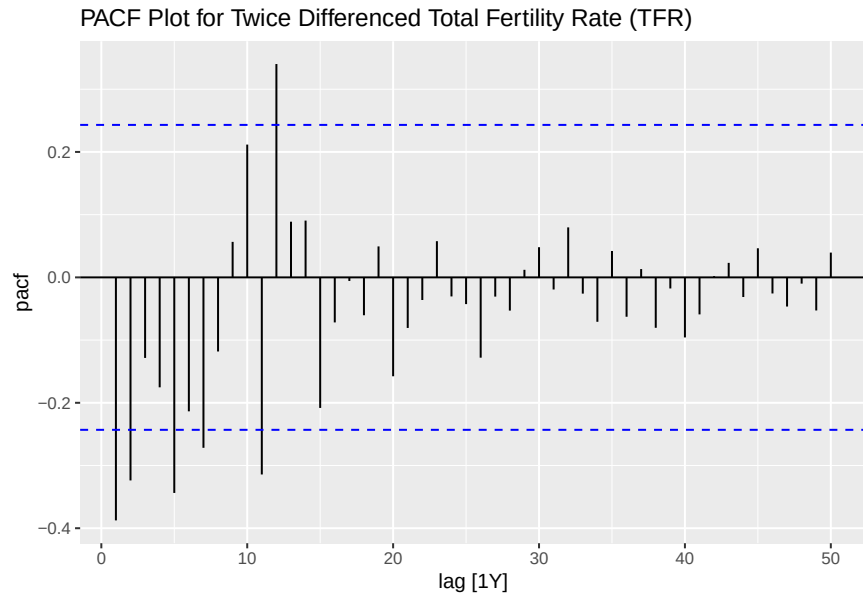
PACF of Residuals for ARIMA(16,1,0) for TFR





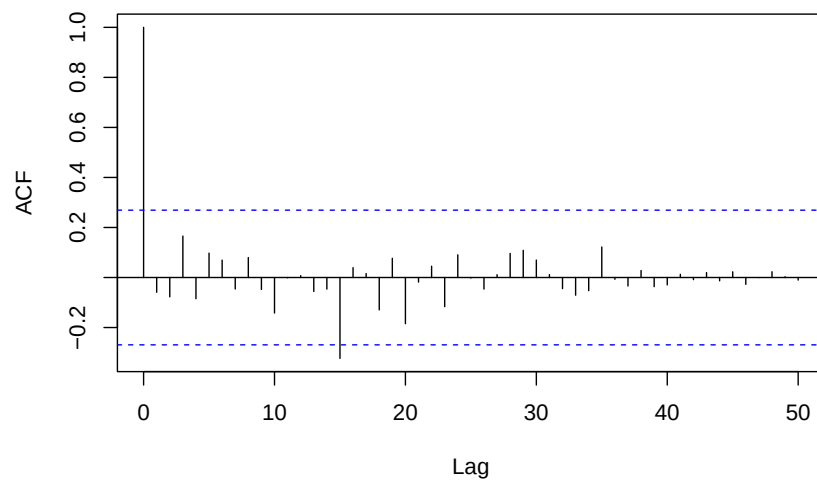
```
##      kpss_stat kpss_pvalue
## 0.04262877 0.10000000
```



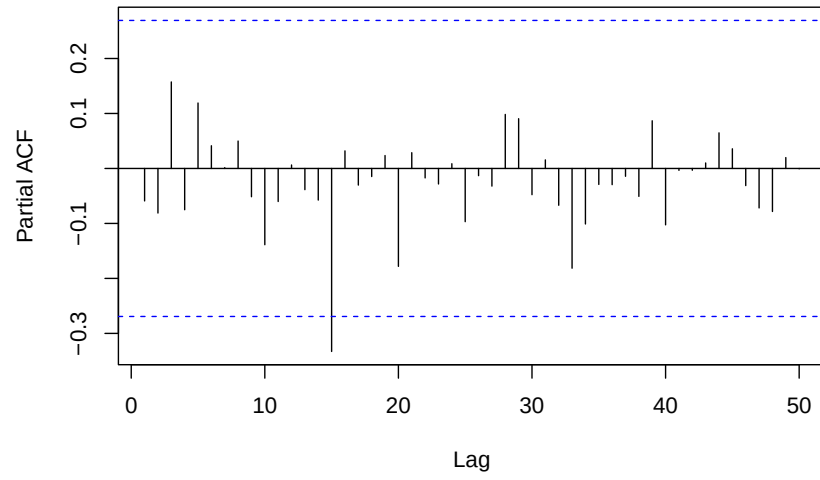


```
## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.6234 -0.7441 -0.5813 -0.5700 -0.4163 -0.1741  0.0462  0.3482
## s.e.   0.1203  0.1494  0.1673  0.1676  0.1705  0.1772  0.1846  0.1847
##      ar9      ar10     ar11     ar12
##      0.4206  0.4880  0.1609  0.5040
## s.e.   0.1845  0.1783  0.1540  0.1205
##
## sigma^2 = 0.01286: log likelihood = 41.78
## AIC=-57.55  AICc=-47.72  BIC=-32.44
```

ACF of Residuals for ARIMA(12,2,0) for TFR



PACF of Residuals for ARIMA(12,2,0) for TFR

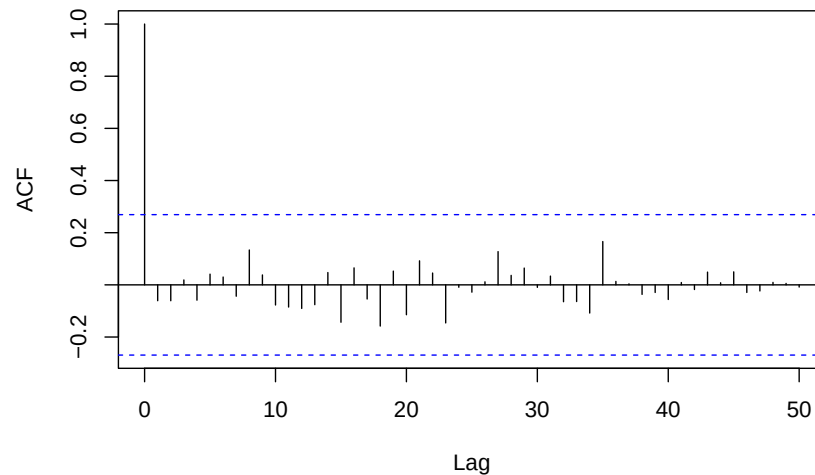



```

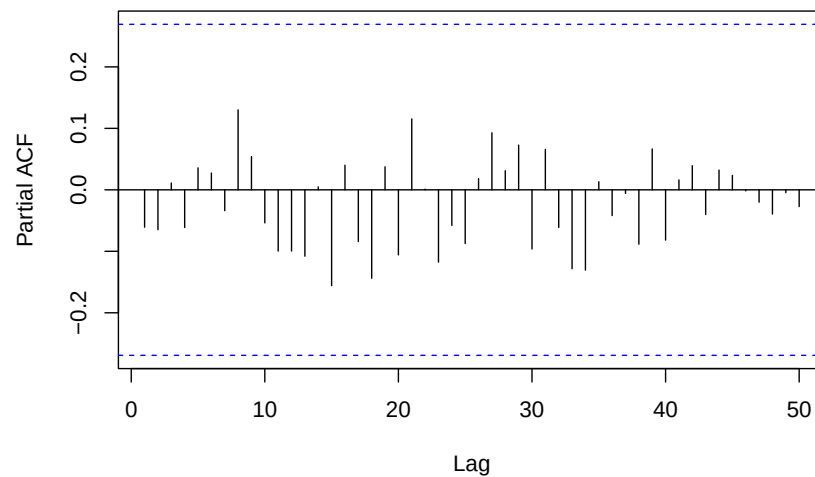
## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.4622 -0.7606 -0.9273 -0.8808 -0.6967 -0.4124 -0.2312  0.0490
## s.e.   0.1091  0.1266  0.1643  0.2032  0.2376  0.2529  0.2548  0.2404
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1578  0.3268  0.1426  0.6463 -0.1402  0.1411  0.8593
## s.e.   0.2067  0.1645  0.1248  0.1005  0.1237  0.1262  0.1309
##
## sigma^2 = 0.009277: log likelihood = 47.29
## AIC=-62.57  AICc=-46.57  BIC=-31.66

```

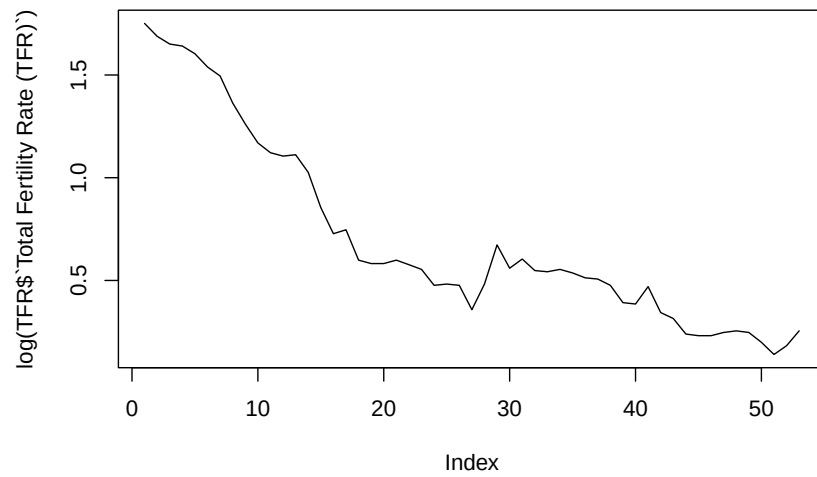
ACF of Residuals for ARIMA(12,2,3) for TFR



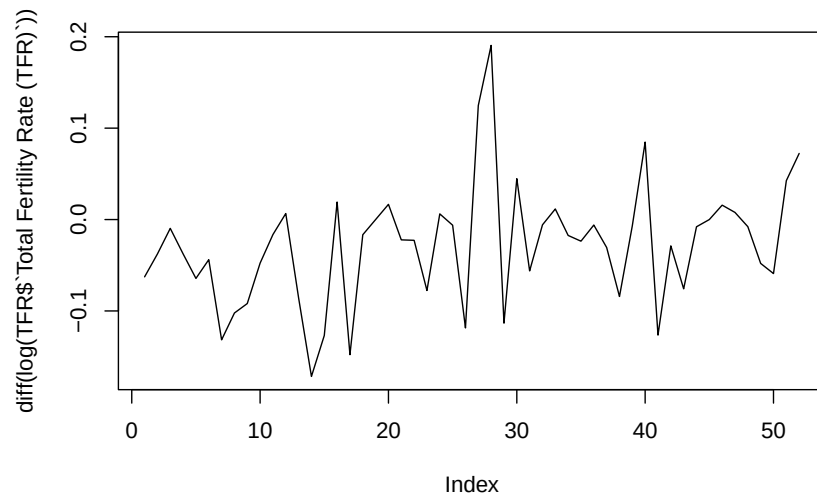
PACF of Residuals for ARIMA(12,2,3) for TFR



Plot of Log Transformation for Total Fertility Rate (TFR) until 2012

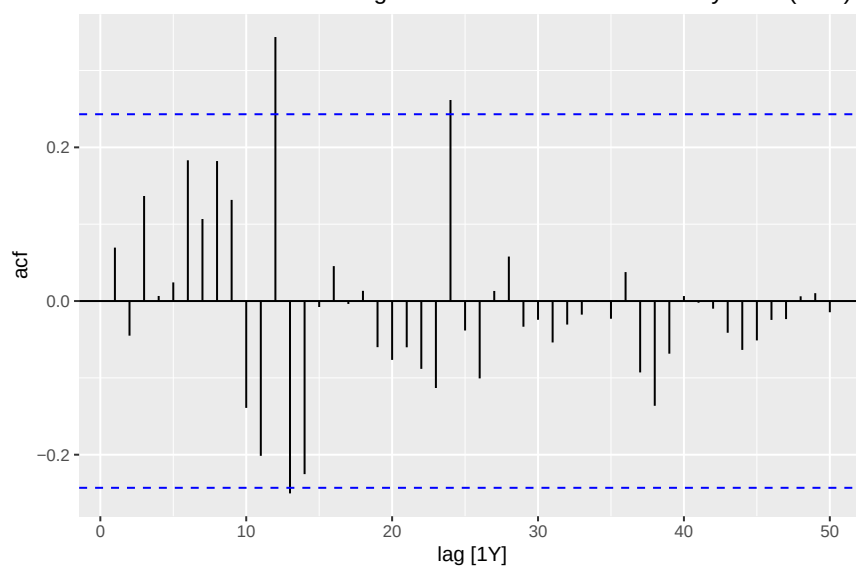


lot of Differenced Log Transformation for Total Fertility Rate (TFR) until

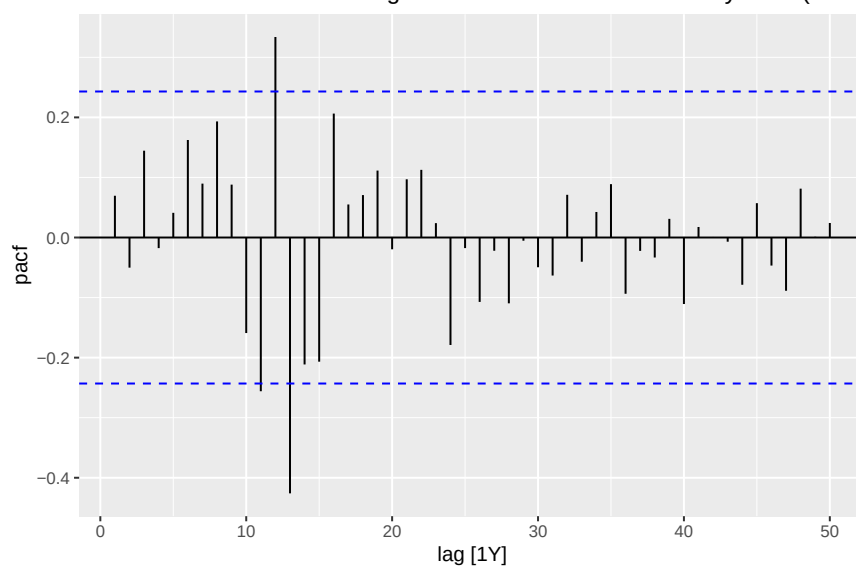


```
##    kpss_stat kpss_pvalue
## 0.50484854 0.04057465
```

ACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

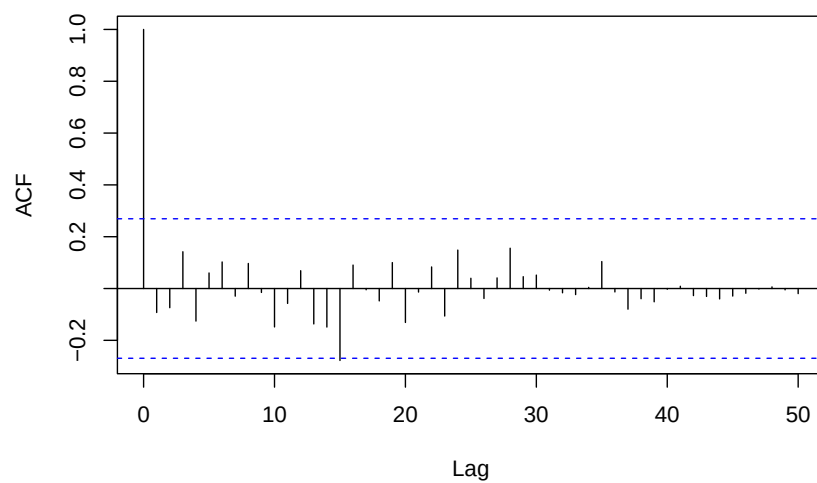


PACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

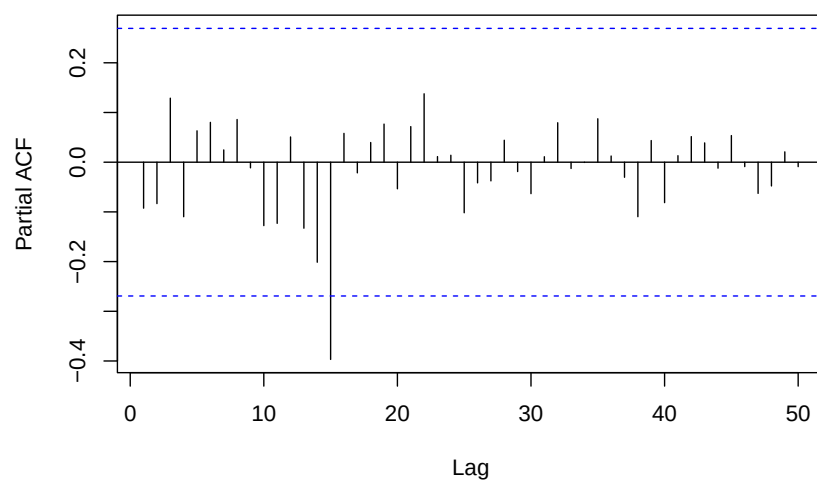


```
## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2629 -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215  0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##      ar10     ar11     ar12     ar13
##      -0.0307 -0.2349  0.3587 -0.4375
## s.e.  0.1130  0.1084  0.1170  0.1258
##
## sigma^2 = 0.00323: log likelihood = 79.9
## AIC=-131.8  AICc=-120.45  BIC=-104.48
```

ACF of Residuals for ARIMA(13,1,0) for TFR with log transformation

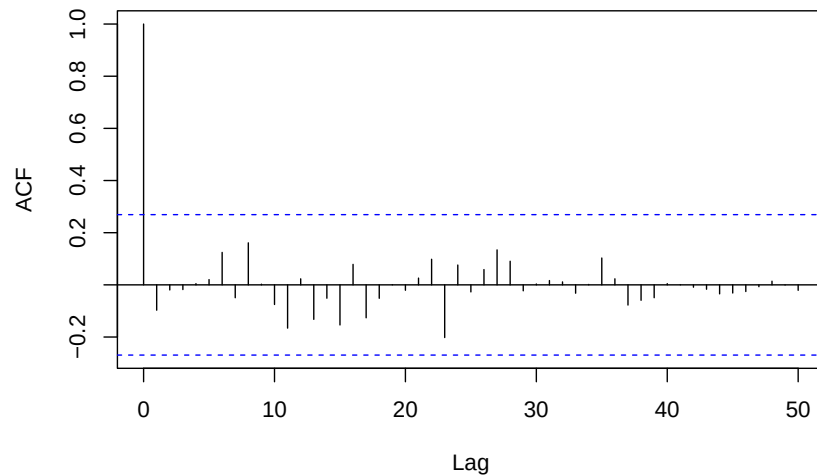


PACF of Residuals for ARIMA(13,1,0) for TFR with log transformation

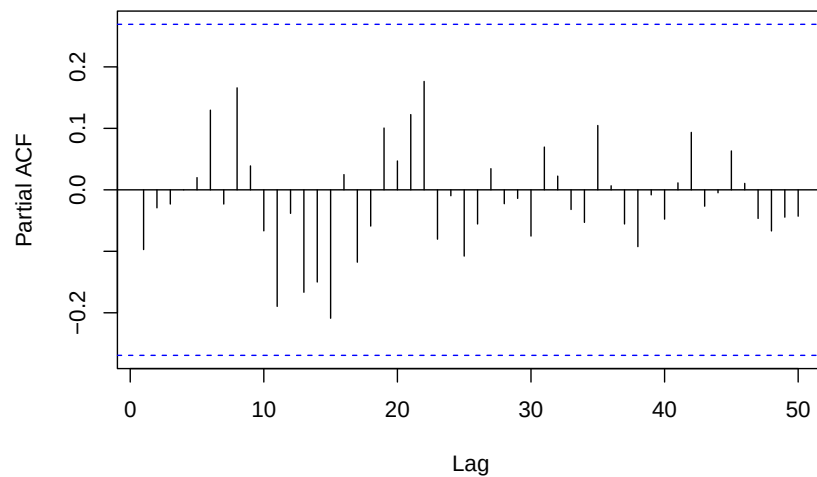


```
## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2783 -0.2434 -0.2961 -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256  0.1019  0.1096  0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##      ar10     ar11     ar12     ar13     ma1      ma2      ma3
##      0.0722 -0.0719  0.4674 -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170  0.1097  0.1005  0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83
```

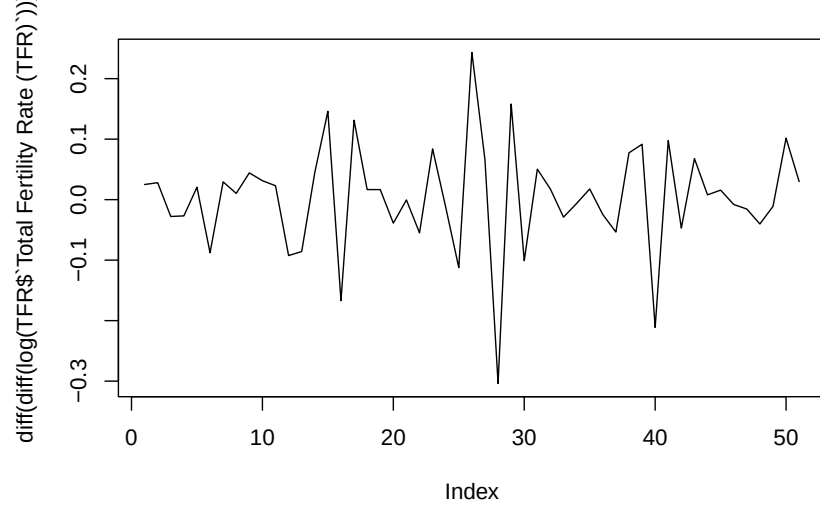
ACF of Residuals for ARIMA(13,1,3) for TFR with log transformation



PACF of Residuals for ARIMA(13,1,3) for TFR with log transformation

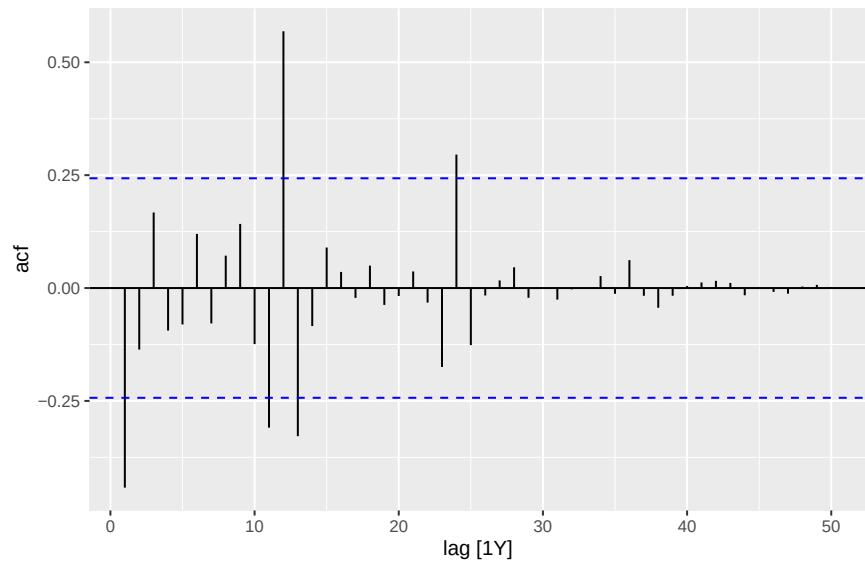


Plot of Differenced Twice Log Transformation for Total Fertility Rate (T

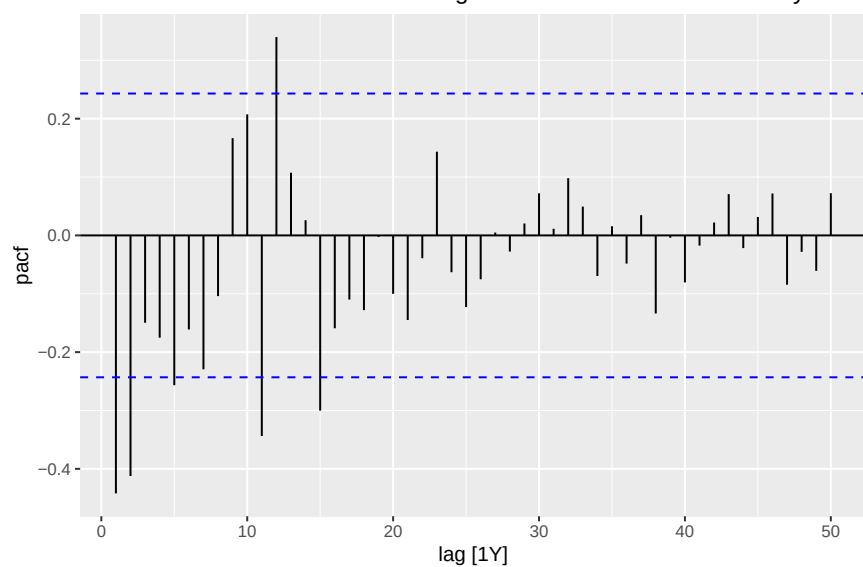


```
##      kpss_stat kpss_pvalue
## 0.04760604 0.10000000
```

ACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (

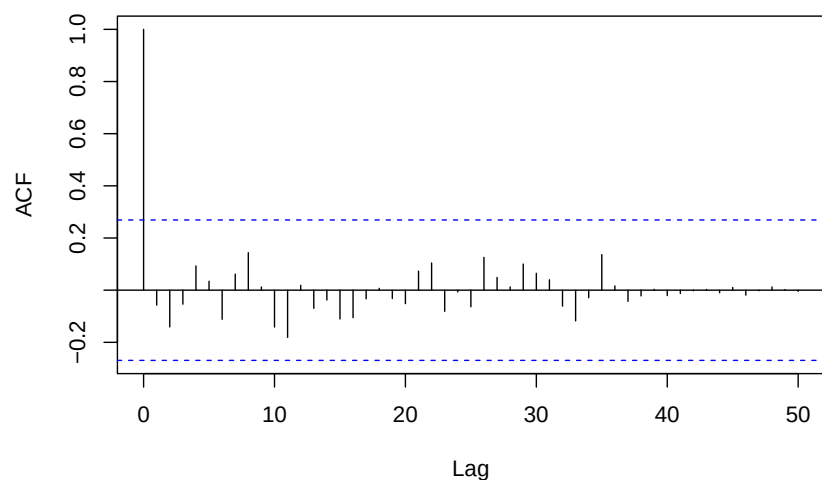


PACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (

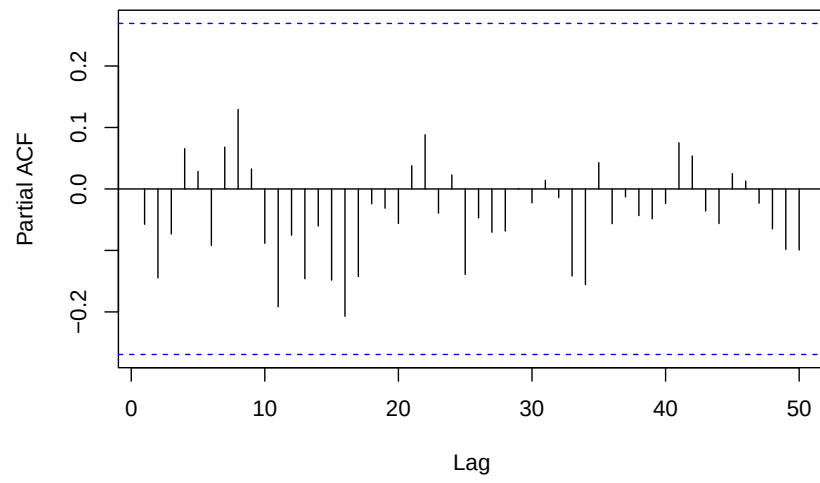


```
## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.7137 -0.7004 -0.3372 -0.5826 -0.3331 -0.1103  0.0628  0.2087
## s.e.   0.1218  0.1571  0.1913  0.1824  0.2031  0.2111  0.2146  0.2149
##      ar9      ar10     ar11     ar12     ar13     ar14     ar15
##      0.1864  0.1142 -0.1162  0.3260 -0.1661 -0.2596 -0.4876
## s.e.   0.2152  0.2137  0.2042  0.1855  0.1901  0.1618  0.1289
##
## sigma^2 = 0.002602: log likelihood = 82.83
## AIC=-133.67  AICc=-117.67  BIC=-102.76
```

ACF of Residuals for ARIMA(15,2,0) for TFR with log transformation



PACF of Residuals for ARIMA(15,2,0) for TFR with log transformatio

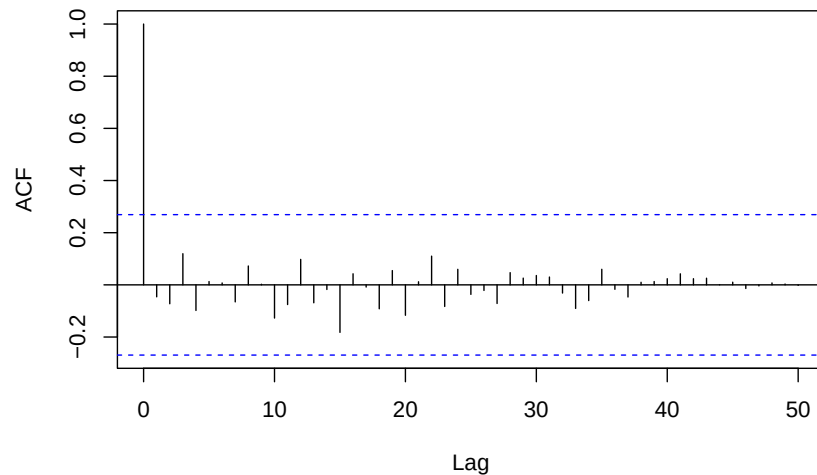



```

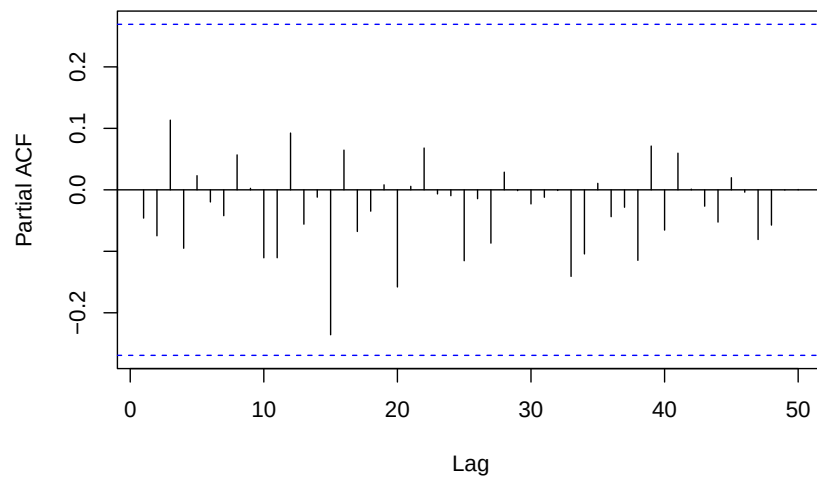
## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6514 -0.2412 -0.0063 -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.  0.1194  0.1202  0.1193  0.1180  0.1158  0.1095  0.1099  0.1125
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.1105 -0.0649 -0.1966  0.4240 -0.6274 -1.6346  0.7851
## s.e.  0.1156  0.1159  0.1124  0.1126  0.0987  0.1446  0.1392
##
## sigma^2 = 0.002717: log likelihood = 82.19
## AIC=-132.39  AICc=-116.39  BIC=-101.48

```

ACF of Residuals for ARIMA(13,2,2) for TFR with log transformation

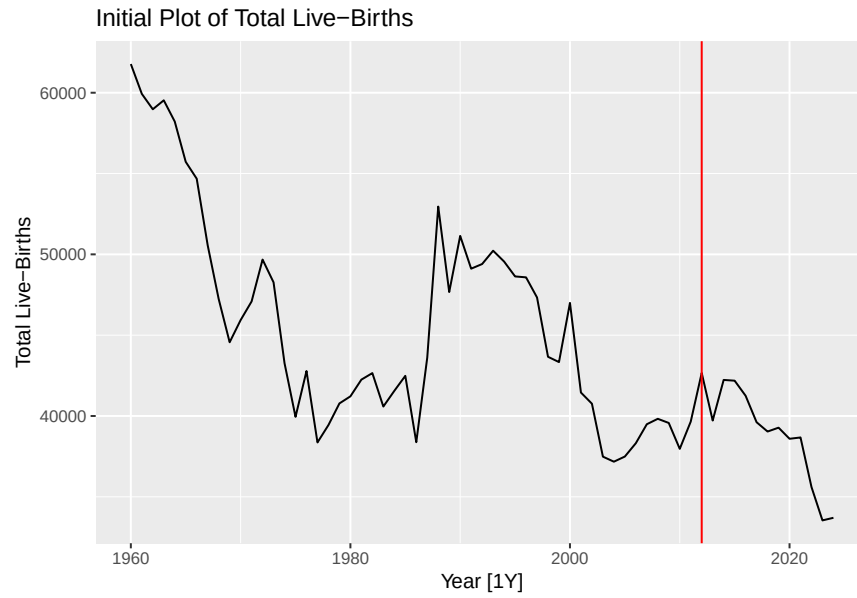


PACF of Residuals for ARIMA(13,2,2) for TFR with log transformation

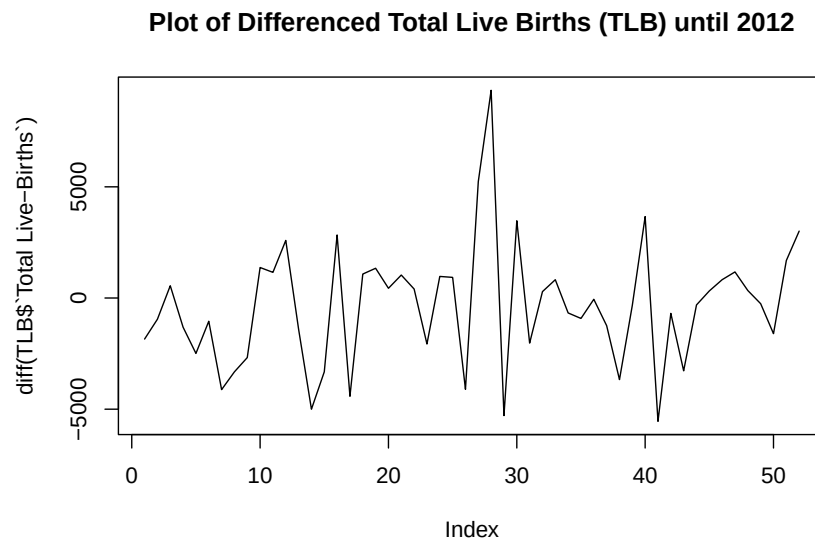


The ACF for TFR shows significant positive autocorrelations from lags 1 to 9, while TLB shows significant positive autocorrelations from lags 1 to 6, with a slow decrease in the ACF as the lag increases. This suggests that neither series is white noise, as there are multiple spikes outside the significance bounds.

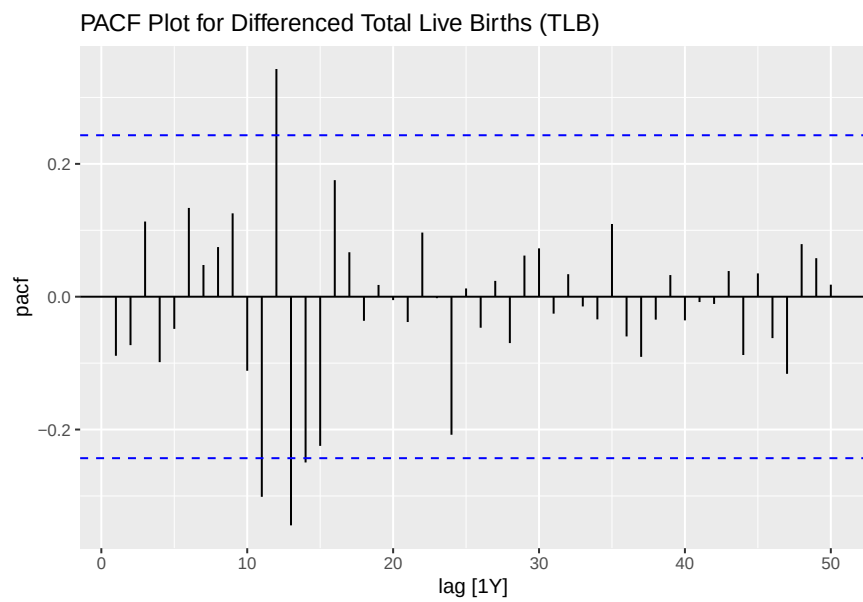
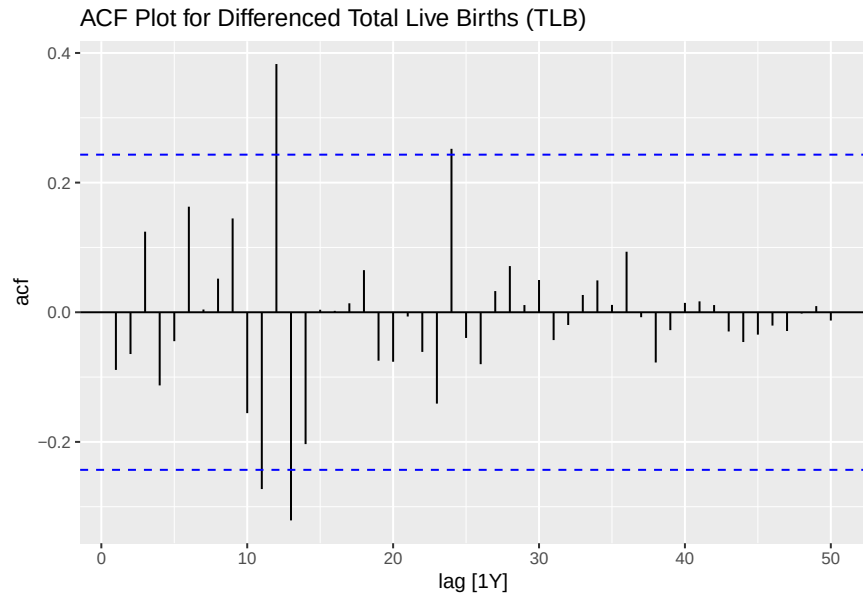
The presence of a strong trend and slow decrease confirms that both TFR and TLB are non-stationary series. The absence of a sharp cut-off suggests that differencing or an autoregressive (AR) model may be suitable for this series. An extended ACF plot for TFR confirms that significant positive autocorrelations persist up to lag 10.



From the initial plots of TFR and TLB, both series show a long-term decreasing trend over time, with TFR exhibiting a more pronounced decline. A notable change is observed around 1980 to 1990, where both series display a temporary broad peak before continuing their downward trend. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclical behaviour, the trend would need to be removed to confirm this.

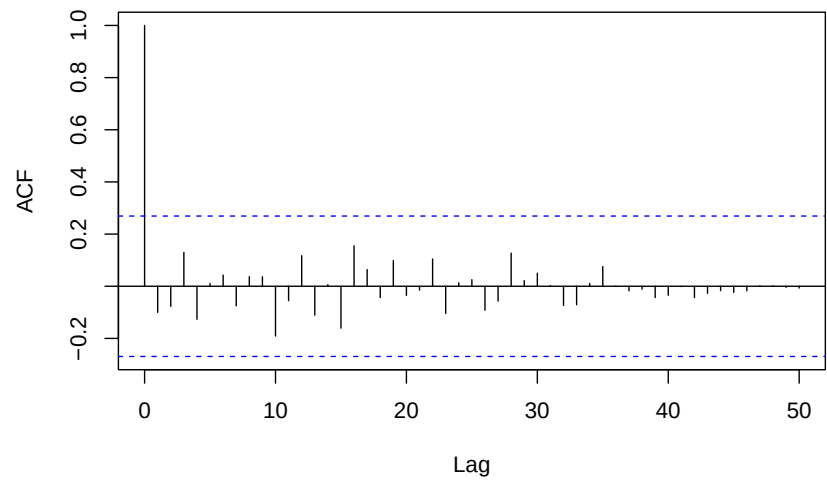


```
##      kpss_stat kpss_pvalue
##      0.2126452  0.1000000
```

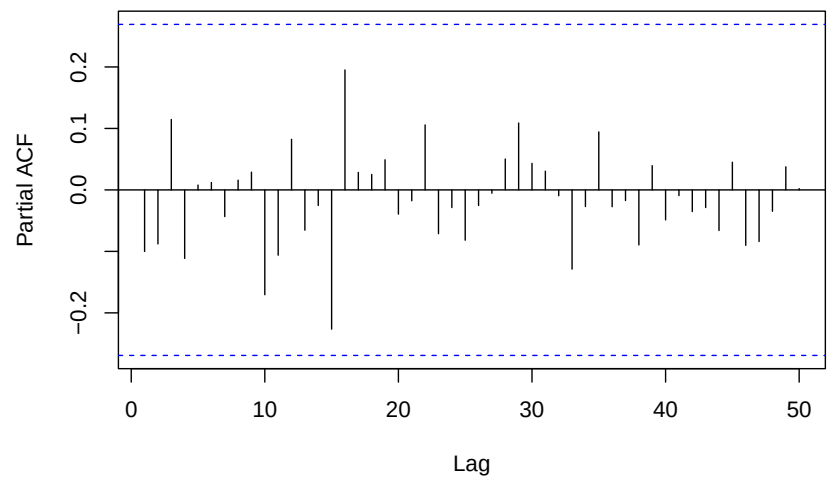


```
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.0260  0.0380 -0.0184 -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.  0.1344  0.1218  0.1091  0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##      ar10     ar11     ar12     ar13     ar14
##      -0.0696 -0.2211  0.2823 -0.3715 -0.2759
## s.e.  0.1085  0.1078  0.1195  0.1245  0.1466
##
## sigma^2 = 5629775: log likelihood = -472.66
## AIC=975.33  AICc=988.66  BIC=1004.6
```

ACF of Residuals for ARIMA(14,1,0) for TLB



PACF of Residuals for ARIMA(14,1,0) for TLB

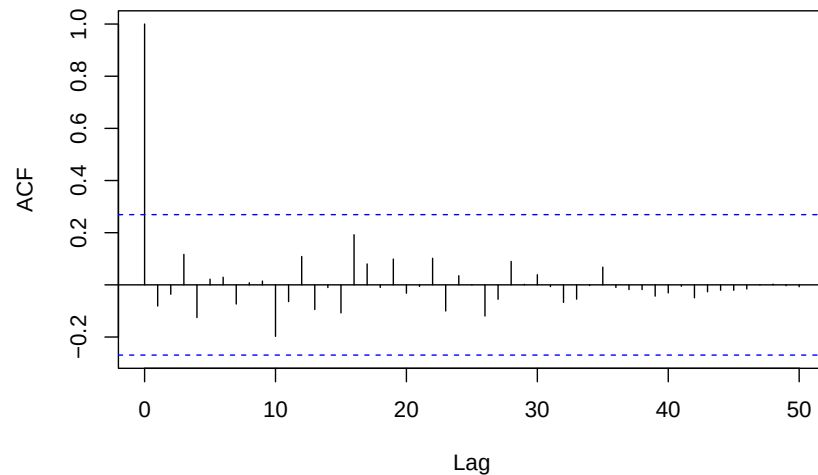


```

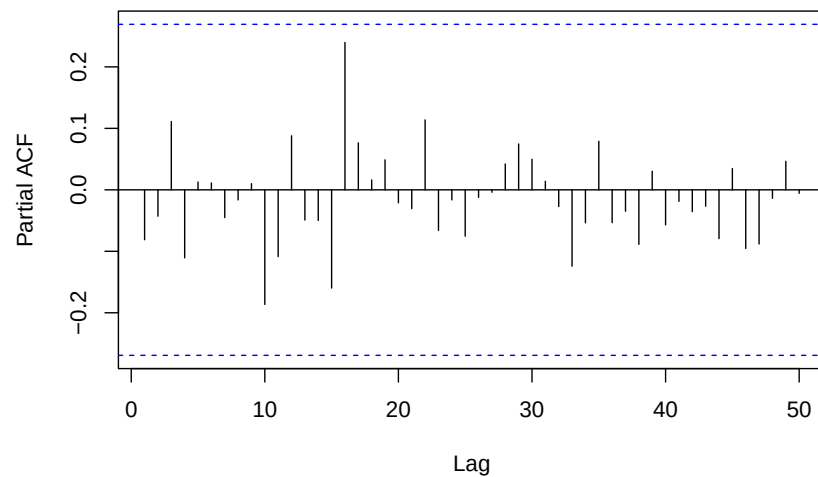
## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.5571 -0.0703  0.0264 -0.0762  0.0940  0.1004  0.0542  0.1077 -0.0216
## s.e.  0.1677  0.1156  0.1116  0.1120  0.1104  0.1040  0.1090  0.1095  0.1125
##      ar10     ar11     ar12     ar13     ma1
##      -0.0853 -0.1964  0.4091 -0.5605 -0.5344
## s.e.  0.1130  0.1112  0.1223  0.1126  0.1886
##
## sigma^2 = 5503186: log likelihood = -472.28
## AIC=974.57  AICc=987.9  BIC=1003.83

```

ACF of Residuals for ARIMA(13,1,1) for TLB



PACF of Residuals for ARIMA(13,1,1) for TLB

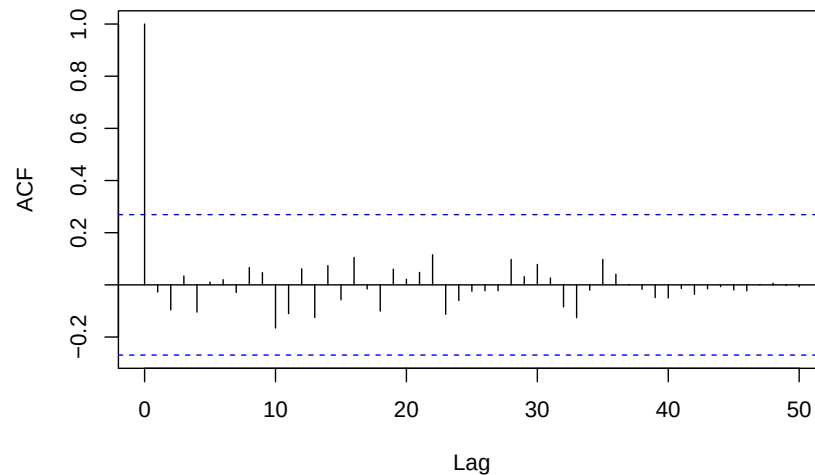


```

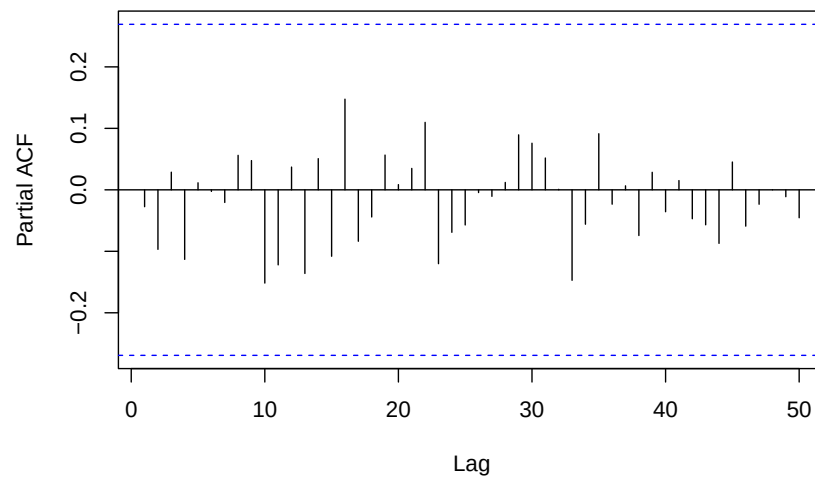
## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6648 -0.4027 -0.0077 -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376  0.1968  0.1260  0.1252  0.1252  0.1203  0.1199  0.1246
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.0053 -0.0478 -0.1305  0.4076 -0.6385 -0.7326  0.5855
## s.e.   0.1262  0.1271  0.1253  0.1159  0.1041  0.1611  0.3171
##
## sigma^2 = 5122169: log likelihood = -470.51
## AIC=973.01  AICc=988.56  BIC=1004.23

```

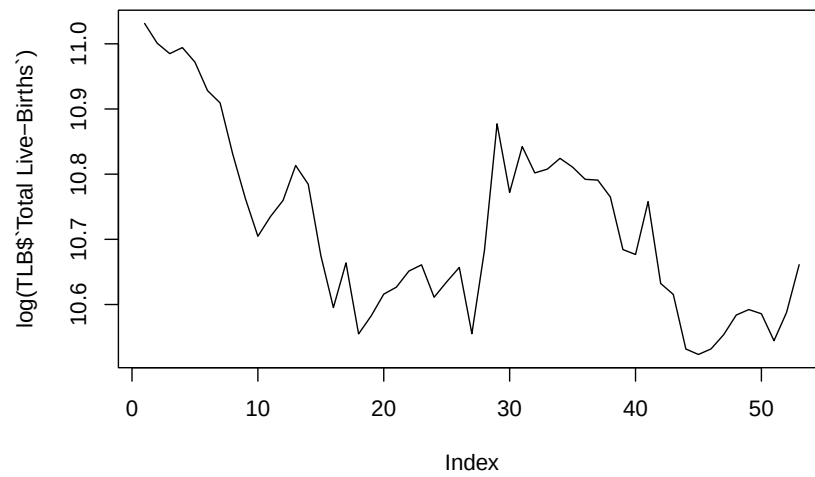
ACF of Residuals for ARIMA(13,1,2) for TLB



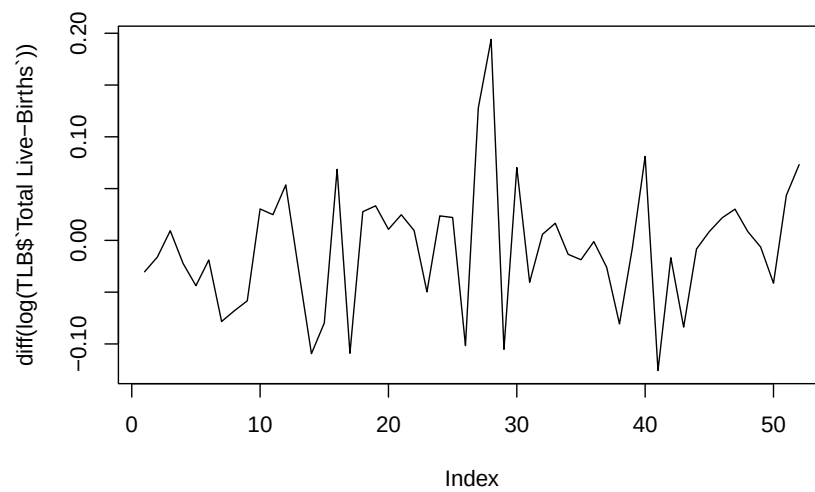
PACF of Residuals for ARIMA(13,1,2) for TLB



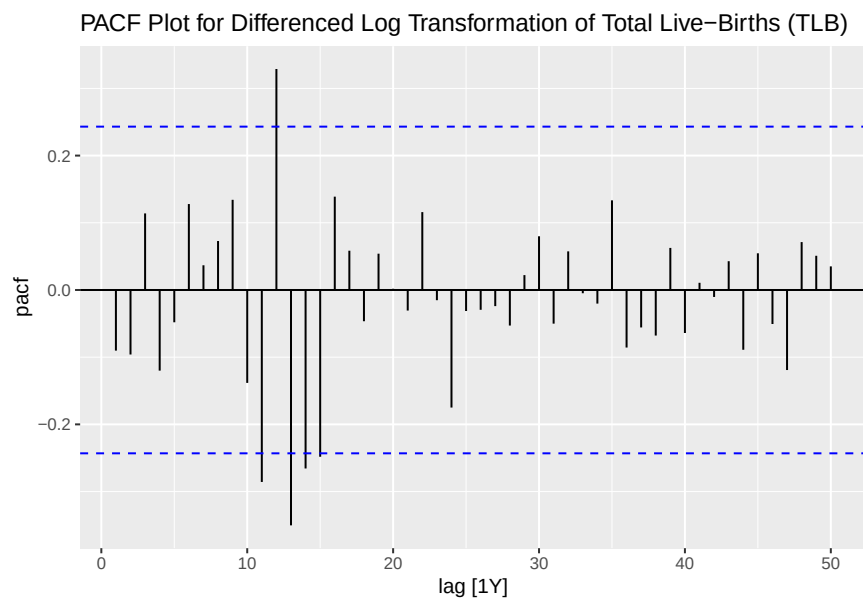
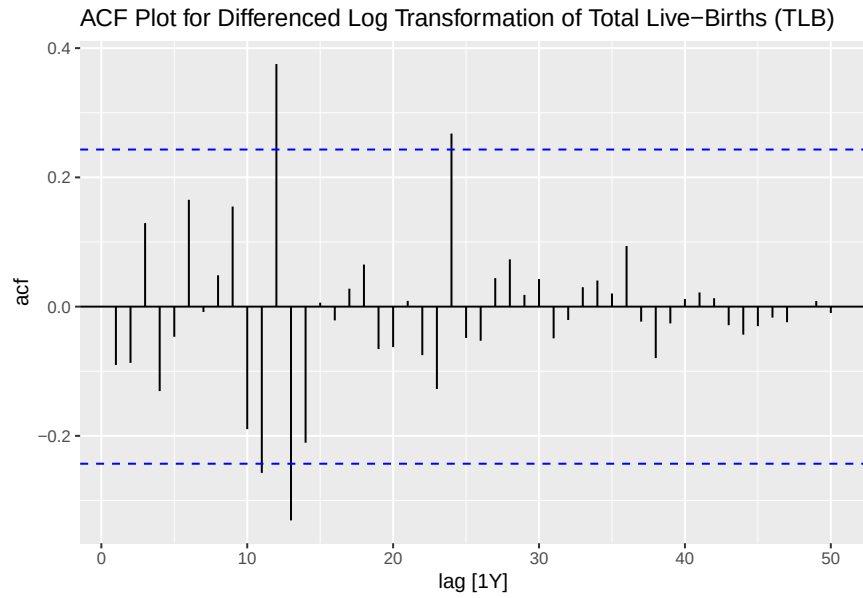
Plot of Log Transformation for Total Live-Births (TLB) until 2012



Plot of Differenced Log Transformation for Total Live-Births (TLB) until

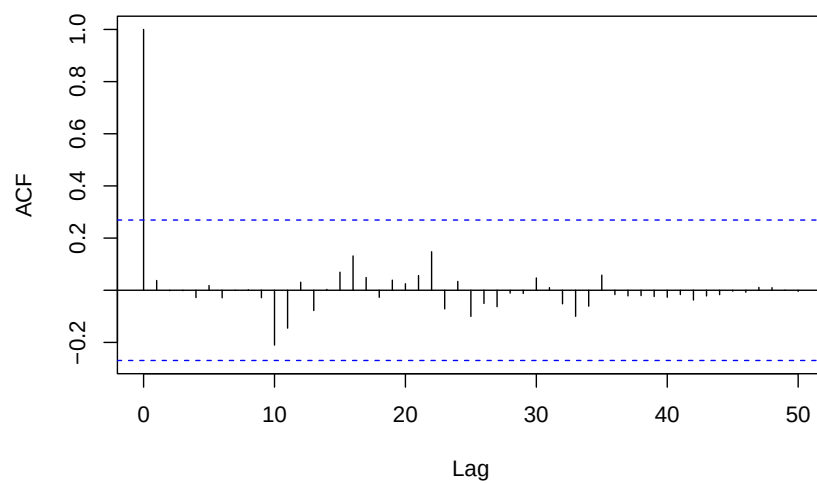


```
##      kpss_stat kpss_pvalue
##      0.1765042  0.1000000
```

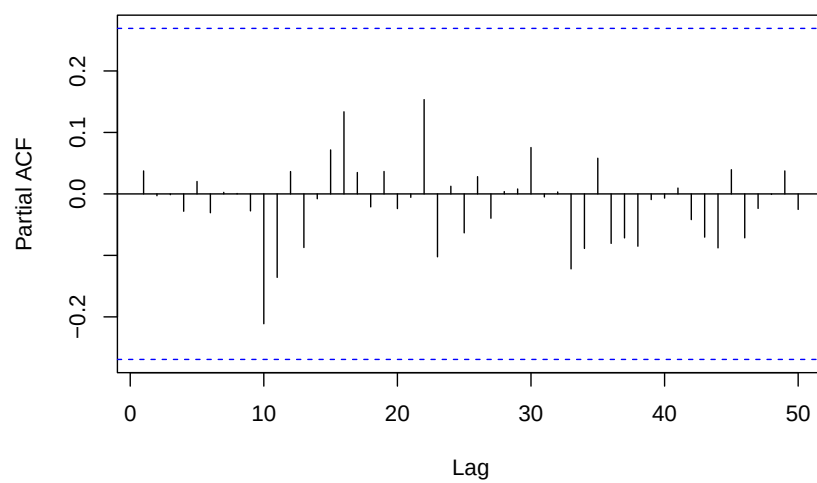



```
## Series: log(TLB_ts)
## ARIMA(15,1,0)
##
## Coefficients:
##          ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      -0.1455 -0.1154  0.0919 -0.1862  0.0220  0.1559  0.1565  0.1742  0.0826
## s.e.   0.1315  0.1197  0.1028  0.0946  0.0935  0.0868  0.0972  0.0918  0.0931
##          ar10     ar11     ar12     ar13     ar14     ar15
##      -0.0706 -0.2325  0.2774 -0.3642 -0.3436 -0.4237
## s.e.   0.0939  0.0892  0.0988  0.1089  0.1264  0.1359
##
## sigma^2 = 0.00225: log likelihood = 88.78
## AIC=-145.56  AICc=-130.02  BIC=-114.34
```

ACF of Residuals for ARIMA(15,1,0) for TLB with log transformation



PACF of Residuals for ARIMA(15,1,0) for TLB with log transformation

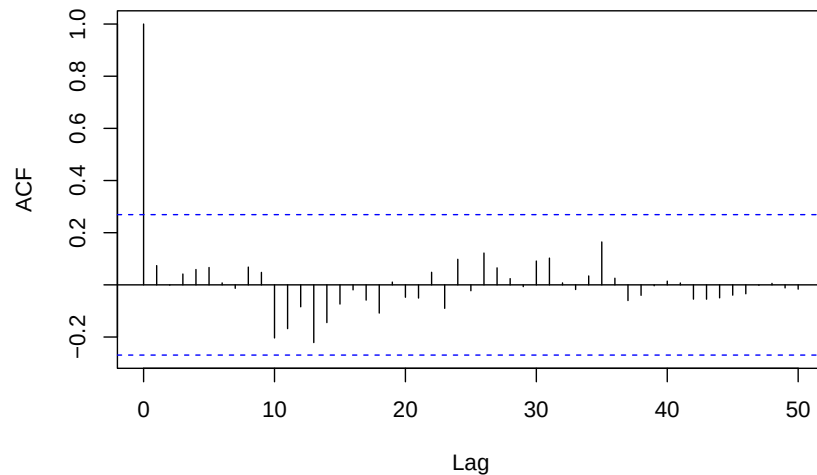


```

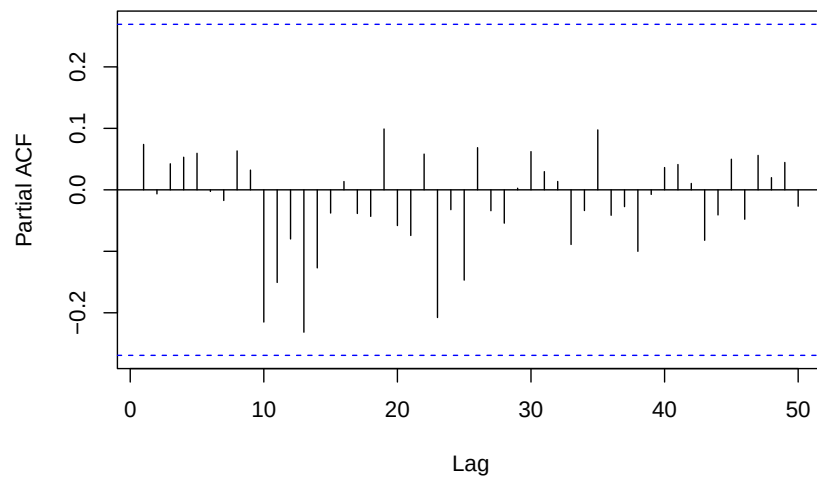
## Series: log(TLB_ts)
## ARIMA(12,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.3816 -0.3915 -0.4801 -0.1650 -0.0459  0.0622  0.0321  0.1705
## s.e.   0.1314   0.1535   0.1615   0.1495   0.1471   0.1454   0.1494   0.1589
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1587  0.0805  -0.0368  0.4897  0.5443  0.4755  0.9169
## s.e.   0.1683  0.1646   0.1479  0.1451  0.1011  0.2255  0.2718
##
## sigma^2 = 0.002424: log likelihood = 85.85
## AIC=-139.69  AICc=-124.15  BIC=-108.47

```

ACF of Residuals for ARIMA(12,1,3) for TLB with log transformation



PACF of Residuals for ARIMA(12,1,3) for TLB with log transformation



Results

TFR Model	AIC	AICc	MSE	MAE
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442
ARIMA(16,1,1)	-62.39	-42.66	0.04783471	0.1700046
ARIMA(15,1,1)	-60.26	-42.26	0.08431368	0.223524
ARIMA(14,1,3)	-60.99	-40.26	0.01899351	0.09964934
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547
ARIMA(11,2,3)	-52.91	-39.19	0.05260825	0.1823268
ARIMA(10,2,3)	-54.69	-43.02	0.05680021	0.1898403

Log Transformed TFR Model	AIC	AICc	MSE	MAE
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592
ARIMA(12,1,4)	-131.83	-113.83	0.02101819	0.1132223
ARIMA(13,1,5)	-133.96	-110.21	0.05826379	0.1840232
ARIMA(12,1,5)	-130.02	-109.3	0.02172126	0.1147922
ARIMA(15,2,0)	-133.67	-117.67	0.01869052	0.1056717
ARIMA(15,2,2)	-135.54	-114.17	0.02235162	0.1187815
ARIMA(14,2,2)	-130.49	-111.94	0.01087075	0.08352761
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933
ARIMA(12,2,2)	-127.34	-113.62	0.0498791	0.1720094

TLB Model	AIC	AICc	MSE	MAE
ARIMA(14,1,0)	975.33	988.66	67367429	6853.008
ARIMA(14,1,1)	975.62	991.17	84179161	7769.997
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802

Log Transformed TLB Model	AIC	AICc	MSE	MAE
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34
ARIMA(15,1,1)	-145.43	-127.43	2173041601	46542.86
ARIMA(14,1,1)	-140.56	-125.02	2211919788	46958.53
ARIMA(13,1,1)	-141.45	-128.11	2200303964	46836.59
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92

Discussion

Conclusion

References

Chen, M., Yip, P.S.F. & Yap, M.T. (2018). Identifying the most Influential Groups in Determining Singapore's Fertility. *Journal of Social Policy*, 47(1), 139-160. doi:10.1017/S0047279417000241

- Goodkind, D. (1996). Chinese Lunar Birth Timing in Singapore: New Concerns for Child Quality amidst Multicultural Modernity. *Journal of Marriage and Family*, 58(3), 784-795. <https://doi.org/10.2307/353736>
- Jones, G.W. & Hamid, W. (2015). Singapore's Pro-natalist Policies: To What Extend Have They Worked? *Low and Lower Fertility: Variations across Developed Countries*, 1, 33-61. Springer. https://doi.org/10.1007/978-3-319-21482-5_3
- Kolambe, M. & Arora, S. (2024). Forecasting the Future: A Comprehensive Review of Time Series Prediction Techniques. *Journal of Electrical Systems*, 20(2), 575-586.
- Newbold, P. (1983). ARIMA Model Building and the Time Series Analysis Approach to Forecasting. *Journal of Forecasting*, 2, 23-35. <https://doi.org/10.1002/for.3980020104>
- Singapore Department of Statistics. (2023). *Births And Fertility Rates, Annual* [Dataset]. data.gov.sg. Accessed March 9, 2026, from https://data.gov.sg/datasets/d_e39eeaeadb571c0d0725ef1eec48d166/view
- Shair, S.N., Shaadan, N., Amirudin, N.A.B.M., & A'kashalf, N.A.M. (2024). Predictions of Malaysia Age-Specific Fertility Rates using the Lee-Carter and the Functional Data Approaches. *AIUB Journal of Science and Engineering*, 23(1), 71-78. doi: 10.53799/ajse.v23i1.1024
- Singh, K., Viegas, O. & Ratnam, S.S. (1988). Fertility Trends in Singapore. *Journal of Biosocial Science*, 20(4), 401-409. <https://doi.org/10.1017/S0021932000017521>
- Tan, J. (2024). Using Decomposed Total Fertility Rate (TFR) to Understand the Drivers for the Decline of TFR in Singapore. *Statistics Singapore Newsletter: Issue 1, 2024*, Household Income and Population Division, Singapore Department of Statistics. <https://www.singstat.gov.sg/files/fc3d9271-e877-4a5c-a197-e244fddd80fb.pdf>
- Vanella, P. (2016). The Total Fertility Rate in Germany until 2040 – A Stochastic Principal Components Projection based on Age-specific Fertility Rates. *Hannover Economic Papers (HEP)*, 579. <https://hdl.handle.net/10419/155634>
- Wilmoth, J., Menozzi, C. & Bassarsky, L. (2022). *Why population growth matters for sustainable development* (Policy Brief No. 130), United Nations Department of Economic and Social Affairs, Population Division. https://www.un.org/development/desa/pd/sites/www.un.org.development.desa.pd/files/undesa_pd_2022_policy_brief_population_growth.pdf
- World Health Organization. *Total fertility rate (per woman)*. Accessed June 7, 2026, from <https://www.who.int/data/gho/indicator-metadata-registry/imr-details/123>

Appendices / Statisal Appendix

Statistical Appendix

The following statistical methods were applied throughout the analysis in the order presented below.

Autoregressive Model (AR)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where:

y_t is the value at time t

ϕ_i are model parameters

ε_t is white noise

Models the current values using its previous values (lags)

Residuals

$$e_t = y_t - \hat{y}_t$$

y_t is the actual observed value at time t

$\hat{y}_t$ is the predicted value at time t

The error which is the difference between the actual observed value and predicted value at time t , used to check the chosen model accounts for all the patterns

Autocorrelation Function (ACF)

$$\rho_k = \frac{\text{Cov}(y_t, y_{t-k})}{\text{Var}(y_t)}$$

where:

ρ_k is the autocorrelation at lag k

Measures the correlation between a series y_t and lagged values y_{t-k} , which checks for trend, seasonality and whether residuals are white noise

Box-Cox Transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

λ was estimated using the Guerrero method

STL Decomposition

$$y_t = T_t + R_t$$

where:

T_t is the trend component,

R_t is the remainder

Decomposes time series into trend and remainder components, usually with the seasonality component as well but the data in this analysis is yearly

Moving Average (MA)

$$MA_k = \frac{1}{k} \sum_{i=-(k-1)/2}^{(k-1)/2} y_{t+i}$$

k is the window size

In this analysis when $k = 5$, the moving average at time t is calculated using the 2 observations before and 2 observations after t . This process smooths out short-term fluctuations in the series

<https://github.com/twalepear/TSA.git>